EDMS 06-300-1

TECHNICAL SPECIFICATION
FOR
SINGLE CORE PVC-INSULATED FLEXIBLE NON SHEATHED CABLES (WIRES) OF RATED VOLTAGES UP TO AND INCLUDING 450/750 V

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Issue: DEC-2021 / Rev- 1

- **The revision contains option items that must be selected by …EDC befor bidding**

CONTENTS

1. SCOPE

This Specification covers the minimum technical requirement for design, engineering, manufacturing, inspection, testing and performance of PVC insulated single core flexible non sheathed cables (wires) rated up to 450/750V, suitable for direct burial, installation in ducts and for general purpose.

2. DEFINITIONS

a) Polyvinyl chloride compound (PVC):

Combination of materials suitably selected, proportioned and treated, of which the characteristic constituent is the elastomer polyvinyl chloride or one of its copolymers. The same term also designates compounds containing both polyvinyl chloride and certain of its polymers

b) Rated voltage:

The rated voltage of a cable is the reference voltage for which the cable is designed and which serves to define the electrical tests

The rated voltage is expressed by the combination of two values Uo/U, expressed in volts:

Uo being the r.m.s. value between any insulated conductor and "earth" (metal covering of the cable or the surrounding medium);

U being the r.m.s. value between any two-phase conductors of a multicore cable or of a system of single-core cables.

In an alternating current system, any rated voltage of a cable shall be at least equal to the nominal voltage of the system for which it is intended

This condition applies both to the value Uo and the value U

Note: The operating voltage of a system may permanently exceed the nominal voltage of such a system by 10%. A cable can be used at a 10% higher operating voltage than its rated voltage if the latter is at least equal to the nominal voltage of the system

c) Test voltage:

Unless otherwise specified, the test voltages shall be a.c. 49 Hz to 61 Hz of approximately sine-wave form, the ratio peak value/r.m.s. value being equal to $\sqrt{2}$ with a tolerance of ±7 %. The values quoted are r.m.s. values.

d) Type tests (symbol T):

Tests required to be made before supplying a type of cable covered by the standard on a general commercial basis in order to demonstrate satisfactory performance characteristics to meet the intended application. These tests are of such a nature that, after they have been made, they need not be repeated unless changes are made in the cable materials or design which might change the performance characteristics.

e) Sample tests (symbol S):

Tests made on samples of completed cable or components taken from a completed cable, adequate to verify that the finished product meets the design specifications.

f) classification:

The conductors have been divided into 4 classes, 1,2,5,6.

Those in class 1 and 2 are intended for use in cables for fixed installations.

Classes 5 and 6 are intended for use in flexible cables and cords but also may be used for fixed installations.

Class 1: solid conductors.

Class 2: stranded conductors.

Class 5: flexible conductors.

Class 6: flexible conductors which are more flexible than class 5.

3. APPLICABLE CODES AND STANDARDS

The latest revision of the following Codes and Standards shall be applicable for the cables covered in this specification. In case of conflict, the vendor/manufacturer may propose equipment/material conforming to one group of Codes and Standards quoted hereunder without jeopardizing the requirements of this Specs:

Table (1)

S.N	Standard No.	Description
1	IEC 60228	Conductors of Insulated Cables.
2	IEC 60811	Electric and optical fiber cables: Test Methods for non-metallic materials.
3	IEC 60227series	Polyvinyl Chloride Insulated Cables Of Rated Voltages Up to And Including 450/750V: Part 1: General Requirements. Part 2: Test Method. Part 3:Non-Sheathed Cables for fixed wiring.
4	ES 2948	Egyptian Standard for Conductors of Insulated Cables.
5	ES 182 Series	Egyptian Standard for Polyvinyl Chloride Insulated Cables Of Rated Voltages Up to And Including 450/750V

4. ENVIRONMENTAL CONDITIONS

The performance of the flexible cables guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Minimum ambient temperature	-5°c
Maximum ambient temperature	45°c (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m.
Maximum conductor temperature in normal use	70°c.

5. DESIGN AND CONSTRUCTION REQUIREMENTS

5.1General

5.1.1 Cables shall meet or exceed the requirements of this Specification in all respects. All cables shall be free of material and manufacturing defects, which would prevent it from meeting the requirements of this Specification.

5.1.2 Manufacturer's drawings shall show the outline of the cables, together with all pertinent dimensions. Any variations in these dimensions due to manufacturing tolerances shall be indicated.

5.2 Design Criteria

5.2.1 Unless otherwise specified, the cables shall be manufactured and tested in accordance with the referred standards.

5.3 Materials

5.3.1 Conductor

The conductor shall consist of copper plain wires, class **5** as per IEC 60228 and ES 2948 and as the following:

Table 3-Flexible copper conductor for single core

Nominal c.s.a mm2	Max diam. Of wires in conductor mm	Max DC resistance of conductor at 20°C
		Plain wires Ω/km
0.5	0.21	39
0.75	0.21	26
1	0.21	19.5
1.5	0.26	13.3
2	0.26	9.89
2.5	0.26	7.98
3	0.31	6.61
4	0.31	4.95
6	0.31	3.30
10	0.41	1.91
16	0.41	1.21
25	0.41	0.780
35	0.41	0.554
50	0.41	0.386
70	0.51	0.272
95	0.51	0.206
120	0.51	0.161
150	0.51	0.129
185	0.51	0.106
240	0.51	0.0801

5.3.2 Insulation

Flexible cables insulation shall be polyvinylchloride (PVC). Insulation shall comply with the appropriate requirements specified in IEC 60227 and ES 182-3.The nominal insulation thickness shall be as per values specified in IEC60227 and ES 182-3, with respect to voltage rating and cables cross-section. The nominal minimum insulation thickness of flexible cables shall be according to relevant standard. The average insulation thickness shall not be less than the specified nominal value. The minimum thickness of the insulation at any point shall not fall below the nominal value by more than 0.1 mm + 10% of the specified nominal value Type of insulation will be PVC/C for single core with no sheath and for flexible cables.

➢ **C.S.A up to 1 mm^2**

1- Rated voltage 300/500V
2- Code designation 60227 IEC06

Table -4 general data for type 60227 IEC06 and ES 182-3

Nominal c.s.a mm2	Thickness of insulation mm	Mean over all diam mm		Min. AC insulation resistance at 70C MΩ/km
		Lower limit	Upper limit	
0.5	0.6	2..1	2.5	0.013
0.75	0.6	2.2	2.7	0.011
1	0.6	2.4	2.8	0.010

➢ **C.S.A ≥ 1.5 mm^2**

1. Rated voltage 450/750V
2. Code designation 60227 IEC02

Table -5 general data for type 60227 IEC02 and ES182-3

Nominal c.s.a mm2	Thickness of insulation mm	Mean over all diam. Mm		Min. AC insulation resistance at 70°C MΩ/km
		Lower limit	Upper limit	
1.5	0.7	2.8	3.4	0.010
2	0.8	3.1	3.7	0.0095
2.5	0.8	3.4	4.1	0.009
3	0.8	3.6	4.4	0.008
4	0.8	3.9	4.8	0.007
6	0.8	4.4	5.3	0.006
10	1	5.7	6.8	0.0056

16	1	6.7	8.1	0.0046
25	1.2	8.4	10.2	0.0044
35	1.2	9.7	11.7	0,0038
50	1.4	11.5	13.9	0.0037
70	1.4	13.2	16	0.0032
95	1.6	15.1	18.2	0.0032
120	1.6	16.7	20.2	0.0029
150	1.8	18.6	22.5	0.0029
185	2	20.6	24.9	0.0029
240	2.2	23.5	28.4	0.0028

6. Marking

All cables shall be marked according to IEC60227-1:

a. The manufacturer's name

b. Voltage designation

c. Type of insulation, PVC

d. Conductor size and material

e. Year of manufacture

f. Cumulative length at every one meter with the highest length mark on the outer end of the cable only for reel.

g. Customer Name ….EDC

- The marking must be repeated at least every 3 m and the length of wire shall be marked every one meter length.
- Color of insulation May be (red – blue – Yellow – Black – Green yellow as required) Optional according to ….EDC requirements

7. TESTS

7.1 General

7.1.1 All cables shall be tested in accordance with the latest standards and as specified herein. Supplier shall provide all test results for review and acceptance by …EDC.

7.1.2 The full range of Sample and Type tests specified in latest version of the relative IEC 60227-3.

7.1.3 Sample tests shall be carried out in the supplier's factory according to table 4 and table 8 of IEC 60227-3. Type test report/certificate shall be submitted to ….EDC.

7.2 Sample Test(S)

Following Sample tests shall be carried out on finished cables:

7.2.1. Electrical Resistance of Conductors, resistance values shall be in accordance with relevant IEC.

7.2.2. AC voltage test, cable shall be tested at test voltage 2500 V For C.S.A$\geq$1.5 mm^2 and 2000 V C.S.A up to 1 mm^2. Values of the test voltages for 5 minutes.

7.2.3. Checking of compliance with constructional previous

7.2.4. Measurement of insulation thickness

7.2.5. Measurement of overall diameter.

7.3 Type Tests:

Table 6-Complete test for C.S.A≥1.5 mm^2 shall be carried out as per IEC 60227-3.

1	2	3	4	
Ref. No.	Test	Category of test	Test method described in	
			IEC	Subclause
1	*Electric test*			
1.1	Resistance of conductors	T, S	60227-2	2.1
1.2	Voltage test at 2 500 V	T, S	60227-2	2.2
1.3	Insulation resistance at 70 °C	T	60227-2	2.4
2	*Provisions covering constructional and dimensional characteristics*		60227-1 and 60227-2	
2.1	Checking of compliance with constructional provisions	T, S	60227-1	Inspection and manual tests
2.2	Measurement of insulation thickness	T, S	60227-2	1.9
2.3	Measurement of overall diameter	T, S	60227-2	1.11
3	*Mechanical properties of insulation*			
3.1	Tensile test before ageing	T	60811-1-1	9.1
3.2	Tensile test after ageing	T	60811-1-2	8.1.3.1
3.3	Loss of mass test	T	60811-3-2	8.1
4	*Pressure test at high temperature*	T	60811-3-1	8.1
5	*Elasticity at low temperature*			
5.1	Bending test for insulation	T	60811-1-4	8.1
5.2	Elongation test for insulation[1)]	T	60811-1-4	8.3
6	*Heat shock test*	T	60811-3-1	9.1
7	*Test of flame retardance*	T	60332-1	

[1)] Only applicable if the overall diameter of the cable exceeds the limits specified in the test method.

Table 7-Complete test for C.S.A (0.5 up to 1) mm^2 shall be carried out as per IEC 60227-3.

1	2	3	4	
Ref. No.	Test	Category of test	Test method described in	
			IEC	Subclause
1	*Electric test*			
1.1	Resistance of conductors	T, S	60227-2	2.1
1.2	Voltage test at 2 000 V	T, S	60227-2	2.2
1.3	Insulation resistance at 70 °C	T	60227-2	2.4
2	*Provisions covering constructional and dimensional characteristics*		60227-1 and 60227-2	
2.1	Checking of compliance with constructional provisions	T, S	60227-1	Inspection and manual tests
2.2	Measurement of insulation thickness	T, S	60227-2	1.9
2.3	Measurement of overall diameter	T, S	60227-2	1.11
3	*Mechanical properties of insulation*			
3.1	Tensile test before ageing	T	60811-1-1	9.1
3.2	Tensile test after ageing	T	60811-1-2	8.1.3.1
3.3	Loss of mass test	T	60811-3-2	8.1
4	*Pressure test at high temperature*	T	60811-3-1	8.1
5	*Elasticity at low temperature*			
5.1	Bending test for insulation	T	60811-1-4	8.1
6	*Heat shock test*	T	60811-3-1	9.1
7	*Test of flame retardance*	T	60332-1	

8. Packing and Shipping

The packing and shipping of the cable shall conform to the following:

8.1. The cable ends shall be sealed with a waterproof, elastomeric end cap for drum

8.2. The cable shall be sealed with heat shrinkable plastic for coils.

8.3. Unless …EDC mention otherwise, Cables shall be supplied in lengths of <u>100</u> meters per roll (up to 25 mm^2) and it should be well sealed against dust, water and weather conditions.

8.4. Unless …EDC mention otherwise, Cables shall be supplied on reels for C.S.A ≥35 mm^2. With the following data shall be printed on both sides:

For ….EDC:

Manufactured by:

Rated voltage: V

C.S.A of the cables: mm^2

Material of conductor: CU

Type of insulation: PVC

Length of cable: M

Cross weight: KGs

Net weight: KGs

Reel No./Date of manufacture:

9. Guarantee

- The supplier should guarantee the Cable against all defects arising out of faulty designed or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, whichever expires earlier.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

10. Drawing and Catalogues:

- All necessary drawings of the cable construction, Technical catalogues containing the technical data and reference standards applied to the cables and instructions for erection, operation, quality control procedure and maintenance should be submitted with the offer.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.
- Type test report/certificate shall be submitted to ….EDC.

11. Guarantee Table

- The tenderer should fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 06-300-1
	Date: 14-12-2021

GUARANTEE TABLE

FOR CU. CABLES…………….. mm^2

-	Manufacturer name	..……
-	Standards to which cables are designed and tested	……
-	Rated voltage	v.
-	Nominal cross-sectional area of conductors	mm^2
-	No of wires / Diameter of each	mm
-	Max. resistance of conductor at 20ºC	Ω/km
-	Insulation thickness	mm
-	Mean overall diameter	mm
-	Min insulation resistance at 70ºC	M Ω .km
-	Weight of CU	Kg/m
-	Gross weight	Kg/m

I/We certify the correctness of the information given above for of cu. cables offered for ….EDC.

Date: / / Signature of Tenderer